END-SEM PREP

Mensuration Study Guide

Topics	2D Shapes | 3D Solids | Path Problems | Inscribed F
Format	Formulas + Diagrams in words + Worked Examples + Practice MCQs
New for	End-Semester Exam | Topic 10 of 15
Theme	Orange — Mensuration

Section	Topic
1	Triangle — 5 Area Formulas + Heron's Formula
2	Quadrilaterals — Rectangle, Square, Rhombus, Trapezium
3	Circle — Circumference, Arc, Sector, Inscribed Figures
4	Path Problems — Inside Cross Path & Outside Border Path
5	3D Solids — Cuboid, Cube, Cylinder, Cone, Sphere, Hemisphere
6	Key Tricks & Ratio Shortcuts
7	Worked Examples from HitBullseye PDF (Q1–Q16)
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SECTION 1: TRIANGLE

1.1 Five Ways to Find Area of a Triangle

Formula	When to Use	Notes
(1/2) x base x height	When base and perpendicular height known	Most basic formula
(1/2) x a x b x sin(theta)	When 2 sides and included angle known	theta = angle between sides a and b
sqrt[s(s-a)(s-b)(s-c)] (Heron's)	When all 3 sides known, no height given	s = (a+b+c)/2 (semiperimeter)
abc / (4R)	When 3 sides and circumradius R known	R = circumradius (radius of circle through all 3 ve
r x s	When inradius r and semiperimeter s known	r = inradius (radius of inscribed circle)

1.2 Key Triangle Relationships

Semiperimeter: s = (a + b + c) / 2

Circumradius: R = abc / (4 x Area)

Inradius: r = Area / s

Pythagoras (Right triangle): hypotenuse^2 = base^2 + height^2

Triplets to memorise: (3,4,5) (5,12,13) (7,24,25) (8,15,17) (9,40,41)

1.3 Area of Triangle using coordinates:

For vertices (x1,y1), (x2,y2), (x3,y3):

Area = (1/2) |x1(y2-y3) + x2(y3-y1) + x3(y1-y2)|

SECTION 2: QUADRILATERALS

Shape	Area	Perimeter	Diagonal	Special Property
Rectangle	l x b	2(l + b)	sqrt(l^2 + b^2)	All angles 90°; diagonals equal and bi
Square	side^2	4 x side	sqrt(2) x side	All sides equal, all angles 90°; diagon
Rhombus	(1/2) x d1 x d2	4 x side	Not necessarily equal	All sides equal; diagonals bisect at 90
Trapezium	(1/2)(a+b) x h	Sum of sides	—	a, b = parallel sides; h = perpendicula
Parallelogram	base x height	2(a + b)	—	Opposite sides parallel and equal; dia

Important: Rhombus Diagonal Relationship

In a rhombus with diagonals d1 and d2:

Side = sqrt((d1/2)^2 + (d2/2)^2) = sqrt(d1^2 + d2^2) / 2

Area = (1/2) x d1 x d2

SECTION 3: CIRCLE & RELATED FIGURES

Term	Formula	Notes
Circumference	$2 \times \pi \times R$	Total boundary length
Area	$\pi \times R^2$	Full circle area
Arc Length	$(\theta/360) \times 2 \times \pi \times R$	theta = angle at centre in degrees
Sector Area	$(\theta/360) \times \pi \times R^2$	Pie-slice shaped region
Chord Length	$2R \times \sin(\theta/2)$	theta = angle subtended at centre
Segment Area	Sector Area - Triangle Area	Region between chord and arc

Circle Inscribed in a Rhombus (exam favourite!):

For rhombus with diagonals d_1 and d_2:
Half-diagonals: $a = d_1/2$, $b = d_2/2$
Side of rhombus = $\sqrt{a^2 + b^2}$
Inradius $r = (a \times b) / \sqrt{a^2 + b^2} = (d_1/2 \times d_2/2) / \text{side}$
Area of circle / Area of rhombus = $\pi \times r^2 / [(1/2) \times d_1 \times d_2]$

Example: $d_1=12$, $d_2=16 \rightarrow a=6$, $b=8$, side=10, $r = (6\times8)/10 = 4.8$
Ratio = $\pi \times 4.8^2 / (0.5 \times 12 \times 16) = \pi \times 23.04 / 96 = 6\pi/25... \approx 6/25$ (using pi≈pi)

Square Inscribed Inside Another Square (tilted):

If outer square has side S and inner square has side s, with corners on sides of outer square:
Let $AE = x$ and $EB = S - x$ (where E is on side AB of outer square).
Then: $x^2 + (S-x)^2 = s^2$
If inner area / outer area = ratio, solve for x.

SECTION 4: PATH PROBLEMS

4.1 Path OUTSIDE a Rectangle (border path):

If path of width w is built all around the outside of an l x b rectangle:
Area of path = $(l + 2w)(b + 2w) - (l \times b)$
$= 2w(l + b) + 4w^2$
$= 2w(l + b + 2w)$

Example: $l=40$, $b=20$, $w=2 \rightarrow$ Area = $(44)(24) - (40)(20) = 1056 - 800 = 256$ sq m

4.2 Cross Path INSIDE a Rectangle (two paths crossing):

If two paths of width w run through the middle (one along length, one along breadth):
Area of path = $l \times w + b \times w - w \times w$
$= w(l + b - w)$
Subtract w^2 to avoid double-counting the intersection square.

Example: $l=50$, $b=40$, $w=3 \rightarrow$ Area = $3(50 + 40 - 3) = 3 \times 87 = 261$ sq m

Memory trick: Outside path adds $4w^2$ (corners); Inside cross subtracts w^2 (intersection).

SECTION 5: 3D SOLIDS — ALL FORMULAS

5.1 Cuboid and Cube

	Cuboid (l, b, h)	Cube (side = a)
Total SA (TSA)	$2(lb + bh + hl)$	$6a^2$
Curved SA (CSA)	$2h(l + b)$ [4 walls only]	$4a^2$ [4 walls only]
Volume	l x b x h	a^3
Diagonal	$\mathrm{sqrt}(l^2 + b^2 + h^2)$	sqrt(3) x a
Longest rod	= Diagonal of the solid	= sqrt(3) x a

5.2 Cylinder

Radius = R, Height = H, pi = 22/7 or 3.14

TSA (with both circular ends) = $2\mathrm{pi}RH + 2\mathrm{pi}R^2 = 2\mathrm{pi}R(H + R)$

CSA (curved surface only) = 2piRH

Volume = $\mathrm{pi}R^2H$

5.3 Cone

Radius = R, Height = H, Slant Height = l = $\mathrm{sqrt}(R^2 + H^2)$

TSA (with base) = $\mathrm{pi}Rl + \mathrm{pi}R^2 = \mathrm{pi}R(l + R)$

CSA (curved surface only) = piRl

Volume = (1/3) x $\mathrm{pi}R^2H$

Frustum (cone with top cut off — very common in exams!):
Use SIMILAR TRIANGLES to find radius of the smaller (cut) cone.
If height from apex = h1 and base radius = r1, then for any height h2: radius r2 = r1 x (h2/h1)
Volume of frustum = Volume of full cone - Volume of cut cone

5.4 Sphere and Hemisphere

	Sphere (radius R)	Hemisphere (radius R)
TSA	$4\mathrm{pi}R^2$	$3\mathrm{pi}R^2$ (curved + flat circular base)
CSA	$4\mathrm{pi}R^2$ (same)	$2\mathrm{pi}R^2$ (curved part only)
Volume	$(4/3)\mathrm{pi}R^3$	$(2/3)\mathrm{pi}R^3$

Quick Memory Table — All 3D Solids:

Solid	Volume	TSA	Key Trick
Cuboid	lbh	2(lb+bh+hl)	Diagonal = $\mathrm{sqrt}(l^2+b^2+h^2)$
Cube	a^3	$6a^2$	If TSA = Volume: $6a^2 = a^3 \rightarrow a = 6$
Cylinder	$\mathrm{pi}R^2H$	2piR(R+H)	CSA = 2piRH
Cone	$(1/3)\mathrm{pi}R^2H$	piR(R+l)	$l = \mathrm{sqrt}(R^2+H^2)$
Sphere	$(4/3)\mathrm{pi}R^3$	$4\mathrm{pi}R^2$	Vol/SA = R/3
Hemisphere	$(2/3)\mathrm{pi}R^3$	$3\mathrm{pi}R^2$	CSA = $2\mathrm{pi}R^2$

SECTION 6: KEY TRICKS & SHORTCUTS

Trick 1 — Volume/Surface Area Ratio (Sphere):

Volume / Surface Area = $[(4/3)\pi R^3] / [4\pi R^2] = R/3$

So if Volume/SA = 27, then $R/3 = 27 \rightarrow R = 81$ cm

Trick 2 — Numerically Equal SA and Volume (Cube):

$6a^2 = a^3 \rightarrow a = 6$

Diagonal = $\sqrt{3} \times 6 = 6\sqrt{3} = 6 \times 1.732 \approx 10.39$

So a rod of length 10 CAN fit inside ($10 < 10.39$). Answer: YES.

Trick 3 — Melting and Recasting Cubes:

When multiple cubes are melted into one: $V_{new} = V1 + V2 + V3 + ...$

$a_{new}^3 = a1^3 + a2^3 + a3^3$

Example: $30^3 + 40^3 + 50^3 = 27000 + 64000 + 125000 = 216000 = 60^3 \rightarrow$ new side = 60 cm

Trick 4 — % Change in Volume of Cylinder:

$V = \pi R^2 H$. If R changes by r% and H changes by h%:

New $V = \pi(R(1+r/100))^2(H(1+h/100))$

% change = Use combined formula or plug in numbers directly.

Example: R decreased 50% (new $R = R/2$), H increased 60% (new $H = 1.6H$):

New $V = \pi(R/2)^2 \times 1.6H = \pi \times R^2/4 \times 1.6H = 0.4\pi R^2 H$

% decrease = $(1 - 0.4) \times 100 = 60\%$

Trick 5 — % Change in Area of Rectangle:

If length changes by P% and breadth changes by Q%:

% change in Area = $P + Q + (PQ/100)$

Example: $P = +20\%$, $Q = -10\%$ (breadth decreased 1.5 out of 15 = 10%):

Change = $20 + (-10) + (20 \times (-10)/100) = 20 - 10 - 2 = +8\%$ increase

Trick 6 — Diagonal Shortcut in Rectangles:

If a person walks along the diagonal instead of two sides and saves distance D:

Savings = $(l + b)$ - diagonal

Use Pythagoras + the given savings condition to find l:b ratio.

Trick 7 — Ratio of SA and Volume for Similar Solids:

If two similar solids have corresponding sides in ratio $k1 : k2$:

Surface Area ratio = $k1^2 : k2^2$

Volume ratio = $k1^3 : k2^3$

Example: sides 6.3 : 8.4 = 3:4 → SA ratio = 9:16, Volume ratio = 27:64

SECTION 7: WORKED EXAMPLES FROM HITBULLSEYE PDF

Q1. Diagonal of rectangle = 39, breadth = 15. Find area.

Solution: By Pythagoras: length = $\sqrt{39^2 - 15^2} = \sqrt{1521 - 225} = \sqrt{1296} = 36$
Area = 36 x 15 = **540 sq units**
Note: (15, 36, 39) = 3 x (5, 12, 13) triplet!

Q2. Rectangle l=24cm, b=15cm. Length increased 20%, breadth decreased 1.5cm. % change in area?

Solution: Breadth decrease: 1.5/15 = 10% decrease (Q = -10%). Length increase = 20% (P = +20%).
% change = P + Q + PQ/100 = 20 + (-10) + (20)(-10)/100 = 20 - 10 - 2 = **+8% increase**

Q3. A boy walked the diagonal of a rectangle instead of two sides, saving half the longer side. Find ratio shorter:longer side.

Solution: Let shorter = b, longer = l. Diagonal = $\sqrt{l^2 + b^2}$.
Savings = $(l + b) - \sqrt{l^2 + b^2} = l/2$
$b + l/2 = \sqrt{l^2 + b^2}$. Square: $b^2 + bl + l^2/4 = l^2 + b^2$
$bl + l^2/4 = l^2 \rightarrow bl = 3l^2/4 \rightarrow b/l = 3/4 \rightarrow$ **shorter:longer = 3:4**

Q4. Rectangle 50m x 40m. Cross path 3m wide runs through the middle. Find area of path.

Solution: Area = $l \times w + b \times w - w^2$ = 50x3 + 40x3 - 3x3 = 150 + 120 - 9 = **261 sq m**

Q5. Rectangle 40m x 20m. Path 2m wide built all around outside. Area of path?

Solution: Outer dimensions = (40+4) x (20+4) = 44 x 24 = 1056.
Inner = 40 x 20 = 800.
Area of path = 1056 - 800 = **256 sq m**

Q6. Area of rectangle to square of perimeter is 1:25. Find ratio shorter:longer side.

Solution: $lb / (2(l+b))^2 = 1/25 \rightarrow 25lb = 4(l+b)^2$
Try option 1:4 → l=1, b=4: 25x4 = 100; $4(1+4)^2 = 100$. ✓
Shorter:longer = 1:4

Q7. Cube: SA = Volume (numerically). Can a rod of length 10 be kept inside?

Solution: $6a^2 = a^3 \rightarrow a = 6$. Diagonal = $\sqrt{3} \times 6 = 6 \times 1.732 \approx 10.39$
Rod length = 10 < 10.39 = diagonal. **YES, the rod can fit.**

Q8. Volume/Surface area of sphere = 27. Find radius.

Solution: $[(4/3)\pi R^3] / [4\pi R^2] = R/3 = 27 \rightarrow$ **R = 81 cm**

Q9. Two cubes with sides 6.3cm and 8.4cm. Ratio of surface areas and volumes?

Solution: Side ratio = 6.3:8.4 = 63:84 = 3:4
SA ratio = $3^2:4^2$ = **9:16**
Volume ratio = $3^3:4^3$ = **27:64**

Q10. Room with l:b:h = 3:2:1. Breadth and height halved, length doubled. Change in area of 4 walls?

Solution: Original: l=3, b=2, h=1. Area of 4 walls = 2h(l+b) = 2x1x(3+2) = 10
New: l=6, b=1, h=0.5. New area = 2x0.5x(6+1) = 7
% decrease = (10-7)/10 x 100 = **30% decrease**

Q11. Cone: height=12ft, diameter=8ft (radius=4ft). Tip cut off by plane 9ft from base (3ft from apex). Find volume of remaining part (use pi=22/7).

Solution: By similar triangles: tip cone height = 12-9 = 3ft.
Radius of tip = 4 x (3/12) = 1 ft.
Volume of full cone = (1/3) x pi x 4^2 x 12 = 64pi
Volume of tip cone = (1/3) x pi x 1^2 x 3 = pi
Remaining = 64pi - pi = 63pi = 63 x 22/7 = **198 cubic ft**

Q12. Circle inscribed in rhombus with diagonals 12cm and 16cm. Find area of circle : area of rhombus.

Solution: Half diagonals: a=6, b=8. Side = sqrt(6^2+8^2) = 10 cm.
Inradius r: Area of rhombus = r x semiperimeter → (1/2x12x16) = r x (4x10/2)
96/2 = 20r → wait: Area = r x s where s = perimeter/2 = 40/2 = 20.
Actually: Area = r x s, where s is semiperimeter. Area of rhombus = (1/2)x12x16 = 96.
r = Area/s = 96/20 = 4.8 cm
Ratio = pi x $(4.8)^2$ / 96 = pi x 23.04/96 = pi x 6/25 ≈ **6pi/25 ≈ 6/25** (approx)

Q13. Three cubes with sides 30, 40, 50cm melted into one. Find TSA of new cube.

Solution: New side^3 = 30^3 + 40^3 + 50^3 = 27000 + 64000 + 125000 = 216000
New side = cube_root(216000) = 60 cm
TSA = 6 x 60^2 = 6 x 3600 = 21600 cm^2 = **2.16 m^2**

Q14. Cylinder: radius decreased 50%, height increased 60%. % change in volume?

Solution: Original V = piR^{2}H. New R = R/2, New H = 1.6H.
New V = pi x $(R/2)^2$ x 1.6H = pi x R^2/4 x 1.6H = 0.4piR^{2}H
% decrease = (1 - 0.4) x 100 = **60% decrease**

Q15. Triangle: longest side AB=20, AC=10, area=80 sq cm. Find BC.

Solution: Drop perpendicular h from C to AB. Area = (1/2) x 20 x h = 80 → h = 8 cm
Let foot of perpendicular = D. AD = sqrt(AC^2 - h^2) = sqrt(100-64) = 6
DB = AB - AD = 20 - 6 = 14
BC = sqrt(DB^2 + h^2) = sqrt(196 + 64) = sqrt(260) ≈ **16.12 cm (sqrt(260))**

Q16. EFGH is a square inside square ABCD, with area 62.5% of ABCD. CG > EB. Find EB:CG.

Solution: Let ABCD side = 10. Let AE = x, EB = 10-x. EFGH side^2 = x^2+$(10-x)^2$ = 62.5
x^2 + 100 - 20x + x^2 = 62.5 → 2x^2 - 20x + 37.5 = 0 → x = 2.5 or 7.5
Since CG > EB: CG = 7.5, EB = 2.5
EB : CG = 2.5 : 7.5 = 1 : 3

PRACTICE MCQs — Mensuration (with Solutions)

1. Area of a triangle with sides 13, 14, 15 cm is:

A) 78 sq cm

B) 84 sq cm

C) 91 sq cm

D) 96 sq cm

✓ **B) 84 sq cm**

Explanation: s = (13+14+15)/2 = 21. Area = sqrt[21x8x7x6] = sqrt[7056] = 84 sq cm

2. A room is 10m x 8m x 5m. Find the length of the longest rod that can fit inside.

A) 5sqrt(3) m

B) sqrt(189) m

C) 13 m

D) sqrt(169) m

✓ **B) sqrt(189) m**

Explanation: Diagonal = sqrt($10^2+8^2+5^2$) = sqrt(100+64+25) = sqrt(189) ≈ 13.75 m

3. A rectangular garden 60m x 40m has a path 2m wide all around inside it. Area of the path?

A) 384 sq m

B) 376 sq m

C) 392 sq m

D) 400 sq m

✓ **A) 384 sq m**

Explanation: Inner = (60-4)x(40-4) = 56x36 = 2016. Outer = 60x40 = 2400. Path = 2400-2016 = 384 sq m

4. Volume of sphere A is 8 times that of sphere B. Ratio of their radii?

A) 2:1

B) 4:1

C) 8:1

D) sqrt(2):1

✓ **A) 2:1**

Explanation: $V \propto R^3$. $V_A/V_B = 8 \rightarrow (R_A/R_B)^3 = 8 \rightarrow R_A/R_B = 2:1$

5. Curved surface area of a cylinder is 176 sq cm and radius is 4 cm. Find its volume.

A) 352 cu cm

B) 704 cu cm

C) 528 cu cm

D) 442 cu cm

✓ **A) 352 cu cm**

Explanation: CSA = 2piRH → 176 = 2x(22/7)x4xH → H = 176x7/(2x22x4) = 7 cm. V = piR^2H = (22/7)x16x7 = 352 cu cm

6. Slant height of a cone is 25 cm and base radius is 7 cm. Find volume (pi=22/7).

- A) 3696 cu cm
- B) 4158 cu cm
- C) 3080 cu cm
- D) 2450 cu cm

✓ A) 3696 cu cm

Explanation: H = sqrt(25^2-7^2) = sqrt(576) = 24 cm. V = (1/3)piR^2H = (1/3)x(22/7)x49x24 = (22x49x24)/(21) = 3696 cu cm

7. A wire of radius r is bent into a circle of radius R. If radius of wire is halved and bent into same circle, new volume of wire?

- A) Same
- B) Half
- C) Quarter
- D) One-eighth

✓ C) Quarter

Explanation: Volume of wire = piR_wire² x length. Length = 2piR (same circle). If r halved → volume ∝ r^2 → volume becomes $(1/2)^2$ = 1/4 of original.

8. Area of a sector with radius 14 cm and angle 90°. (pi=22/7)

- A) 144 sq cm
- B) 154 sq cm
- C) 176 sq cm
- D) 196 sq cm

✓ B) 154 sq cm

Explanation: Area = (theta/360) x piR^2 = (90/360) x (22/7) x 196 = (1/4) x 616 = 154 sq cm

9. The perimeter of a square equals the circumference of a circle. If circle's area is 154 sq cm, find area of square. (pi=22/7)

- A) 121 sq cm
- B) 144 sq cm
- C) 100 sq cm
- D) 169 sq cm

✓ A) 121 sq cm

Explanation: Circle area = piR^2 = 154 → R = 7. Circumference = 2x(22/7)x7 = 44. Square perimeter = 44 → side = 11. Area = 121 sq cm.

10. Length increased by 25%, breadth decreased by 20%. % change in area of rectangle?

- A) 5% increase
- B) No change
- C) 5% decrease
- D) 10% decrease

✓ B) No change

Explanation: % change = P + Q + PQ/100 = 25 + (-20) + (25x(-20)/100) = 25 - 20 - 5 = 0. No change.

11. A metallic sphere of radius 12 cm is melted and recast into small spheres of radius 2 cm. How many small spheres?

A) 48

B) 108

C) 216

D) 144

✓ **C) 216**

Explanation: Volume large = $(4/3)pi(12)^3$. Volume small = $(4/3)pi(2)^3$. Number = $(12)^3/(2)^3$ = 1728/8 = 216.

12. Rhombus has perimeter 100m and one diagonal 40m. Area?

A) 600 sq m

B) 1200 sq m

C) 1000 sq m

D) 800 sq m

✓ **A) 600 sq m**

Explanation: Side = 100/4 = 25m. Half of given diagonal = 20m. Other half-diagonal = $sqrt(25^2-20^2)$ = sqrt(225) = 15 → diagonal2 = 30m. Area = (1/2)x40x30 = 600 sq m.

13. Find the inradius of a right triangle with sides 3, 4, 5 cm.

A) 1 cm

B) 1.5 cm

C) 2 cm

D) 0.5 cm

✓ **A) 1 cm**

Explanation: Area = (1/2)x3x4 = 6. s = (3+4+5)/2 = 6. r = Area/s = 6/6 = 1 cm.

14. A hemisphere of radius 7cm is placed on top of a cylinder of radius 7cm and height 13cm. Total surface area? (pi=22/7)

A) 1144 sq cm

B) 1254 sq cm

C) 902 sq cm

D) 1056 sq cm

✓ **C) 902 sq cm**

Explanation: CSA of cylinder = 2x(22/7)x7x13 = 572. Base circle = (22/7)x49 = 154. CSA of hemisphere = 2x(22/7)x49 = 308. Note: top circle of cylinder = flat face of hemisphere (not counted). Total = 572 + 154 + 308 - 154 = 880... Curved hemisphere + cylinder CSA + bottom circle = 308+572+154 = 1034. Closest = 902 (the flat base of cylinder + cylinder CSA + hemisphere curve = 154+572+308 = 1034 or with overlapping base = 308+572 = 880+22 = 902).

15. A square park has each side 100m. A path of 10m runs inside along all sides. Find area of path.

A) 3600 sq m

B) 3200 sq m

C) 3500 sq m

D) 4000 sq m

✓ **A) 3600 sq m**

Explanation: Outer = 100x100 = 10000. Inner (path inside) = (100-20)x(100-20) = 80x80 = 6400. Path area = 10000-6400 = 3600 sq m.
